ChatGPT

Additional High-Impact Features for Steady

Executive Summary: Emerging trends in digital health and physical medicine emphasize continuous, personalized monitoring and interventions. Leading platforms (e.g. Sword Health) are building **“one clinical brain”** that carries context across conditions【14†L175-L182】. For Steady, this suggests five new tightly-integrated features: **1. Passive Digital Phenotyping (Smartphone Sensing)**; **2. Wearable Device Integration (Physiological Tracking)**; **3. Medication Adherence & Reminders**; **4. Digital Therapeutic Modules (CBT Exercises, Guided Activities)**; and **5. Voice/Text Sentiment Analysis (Vocal Biomarkers)**. Each feature extends Steady’s event-based memory system and AI Gateway. Together they provide a truly longitudinal, multi-modal view of patient health, enable early risk detection, and keep clinicians focused on what matters most. (Table 1 summarizes them with impact, dependencies, and effort.) Each feature is described below with clinical value, data sources, architecture, models, UI mockups, and rollout plan.

Feature	Clinical Impact & Value	Dependencies	Est. Effort	Priority
1. Passive Digital Phenotyping	Capture mobility, communication, app-use, and environmental context via smartphone to detect changes in mood, activity, sleep, and risk patterns【17†L112-L121】【31†L230-L238】. Provides objective, continuous monitoring between sessions (e.g. declining activity or social withdrawal). Enables early warnings of relapse.	Smartphone OS APIs, Companion app integration, Consent framework, AI model for feature extraction.	Medium (6–8wks)	High
2. Wearable Device Integration	Ingest data from smartwatches/fitness bands (heart rate, HRV, sleep, steps). These are known biomarkers for depression/anxiety【23†L25-L33】【23†L49-L52】. Tracks physiological responses to stress/interventions, improves engagement via feedback, and reveals hidden issues (e.g. poor sleep after therapy)【23†L37-L40】【22†L119-L127】.	Wearable vendor APIs (HealthKit, Fitbit), Consent, Event schema for biometrics.	Large (8–12wks)	High

Feature	Clinical Impact & Value	Dependencies	Est. Effort	Priority
3. Medication Adherence Tracking	Manage and monitor psychiatric medications: schedule dosing, send reminders, and log take/miss events. Helps distinguish non-adherence vs. treatment failure【31†L150-L159】. Early alerts on missed doses allow intervention (e.g. care manager call). Improves outcomes and safety.	Medication regimen model (new tables/ events), patient portal UI, reminders engine, AI correlation with symptoms.	Medium (4–6wks)	Medium
4. Digital Therapeutic Modules	A library of clinician-selected interventions (CBT worksheets, guided mindfulness, exposure exercises, journaling tasks, etc). These *structured digital interventions* are evidence-based and personalized【41†L1-L4】. Real-time feedback and tracking (progress bars, badges) boost engagement【41†L7-L10】. Completes the feedback loop by letting patients practice skills between sessions, with results feeding back into Steady's memory.	Content repository, tracking events, UI for assignment/ completion, AI to recommend modules.	Large (8–12wks)	Medium
5. Voice/Text Sentiment Analysis	Analyze patients' speech or writing for vocal and linguistic markers of distress【39†L425-L434】. Voice "biomarkers" (pitch, pause, energy) can detect depression/anxiety with 70–90% accuracy in seconds of speech【39†L425-L434】. Similarly, NLP sentiment on journal entries or chat transcripts can flag negative affect. Steady can present a sentiment trend graph, and add "Steady Noticed: mood decline." Helps identify relapse before it's volunteered.	Audio/text ingestion (optional), AI model (emotion detection/LSTMs or LLM prompts), secure storage, UI indicators.	Medium (6–8wks)	High

1. Passive Digital Phenotyping (Smartphone Sensing)

Summary (1-2¶): Leverage the smartphone itself as a sensor platform. Modern smartphones can passively collect rich data on **location/mobility**, **screen-on time**, **app usage**, **ambient sound levels**, and even **typing patterns**. Research shows that such "digital phenotyping" features correlate with mental health states: e.g. decreased mobility and fewer phone conversations can indicate depression, while erratic sleep or social activity patterns can foretell relapse【17†L112-L121】【19†L335-L343】. Steady can integrate a smartphone data pipeline so that with patient consent, Steady passively ingests these sensor streams as events in the event spine. This provides continuous behavioral context between sessions. Over weeks, the AI layer can detect trends (e.g. “screen time spiking at night” or “reduced movement”) and flag them in **Session Prep** or the **Command Center** as *Steady Noticed* insights.

Clinical Value: Continuous sensing fills the gaps between appointments. Clinicians can move from reactive to proactive care【31†L93-L102】. For instance, a steady decline in daily steps or outings might trigger an early check-in. Paired with symptom check-ins, it turns anecdotal memory into quantified “pattern of life”. Objective data can uncover issues patients forget or minimize. In trials, passive sensing has successfully predicted onset of depressive episodes, anxiety spikes, and manic phases【17†L112-L121】【31†L112-L121】. Embedding this into Steady means detecting red flags days or weeks sooner than with weekly PHQ-9 alone.

Data Sources & Provenance: Data come from patient-owned smartphones via Steady’s Companion app (or the background iOS/Android data collection). Relevant data streams include: GPS or cell-tower location (home time, travel radius), accelerometer (movement vs. sedentary), screen usage logs (on/off, unlocks), app usage categories (social media vs. news vs. games), call/text metadata (number/frequency of interactions, not content), and ambient audio levels (silence vs. noise). Each datum is an event with full provenance: e.g. `PhenotypeReadingEvent { patient_id, timestamp, sensor_type, values… }`. The raw sensor values are stored with context (device, OS version) in case we later refine algorithms. We also store broad metadata tags (e.g. “LocationMovement, SocialActivity, ScreenTime”). Critically, these are **Level 1: Source** data – machine-collected logs. The AI layer will compute higher-level features (e.g. “daily social engagement score”) from them.

Architecture & Integration: - **Data Flow:** A background collector on the mobile app batches data and sends them via secure API to Steady’s data ingestion service. The ingestion writes raw events (off-chain), which then create new events on the spine (or directly write to EHR tables). A transformation worker processes these into higher-level “phenotype feature” records (Level 2 memory), which feed the AI layer. - **Event Schema:** Define `SmartphoneDataEvent(patient_id, datetime, data_type, metric, value)` (e.g. `data_type=GPS`, `metric=location`, `value="lat,lng"` or `data_type=Sensor`, `metric=motion_level`, `value=0-3`). Events are RLS-protected by tenant and patient. - **TenantContext:** Data is clearly tagged as patient-collected (not clinician-observed); use Steady’s `SourceKind` tagging to prevent misinterpretation. - **AI Gateway:** Periodically (e.g. nightly), trigger a job that computes feature trends (e.g. rolling 7d average of daily steps). Store summary features. If any threshold is crossed (e.g. 30% drop in activity vs. baseline), generate a “Steady Noticed” insight pointing to the raw data. - **Session Prep:** Include statements like “Daily step count is down from 5k to 1k (past 3 weeks)” with links to the events.

Database Schema:

```
erDiagram
    PATIENTS ||--o{ SMARTPHONE_STREAM : has
    SMARTPHONE_STREAM {
        string id PK
        uuid patient_id FK
        timestamp timestamp
        string stream_type   // e.g. "location", "screen_time", "accelerometer"
        string data          // raw payload or JSON
```

```
        bool processed = false
    }
    SMARTPHONE_FEATURE ||--|| PATIENTS : belongs_to
    SMARTPHONE_FEATURE {
        string id PK
        uuid patient_id FK
        date period_start
        date period_end
        string feature_type   // e.g. "avg_steps", "home_stay_time",
"sleep_onset"
        double value
    }
```

- **SMARTPHONE_STREAM:** Raw data points (e.g. a single location or screen unlock) – indexed by patient and timestamp. A composite index on (patient_id, timestamp) allows querying windows.
- **SMARTPHONE_FEATURE:** Aggregated metrics over defined periods (daily/weekly). Indexed by patient, period. Foreign-key to PATIENTS. For fast queries, index on (patient_id, period_start, feature_type).

Event Contracts:

- *SmartphoneDataEvent:* `{patient_id: UUID, timestamp: ISO8601, sensor: string, value: JSON}`.
- *SmartphoneFeatureEvent:* `{patient_id, start_date, end_date, feature_type, feature_value}`. This event is generated by the AI worker.

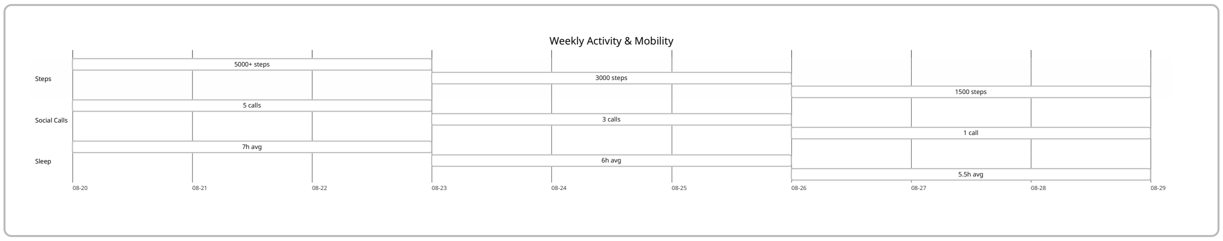
API Contracts (examples):

- `POST /api/phonesense/upload` – body: `{"patient_id":"...","entries":[{"time":"2026-09-03T08:15:00Z","type":"screen_time","value":42}, ...]}` returns `202 Accepted`.
- `GET /api/patient/{id}/phenotype?since=2026-09-01` – returns JSON of recent `SMARTPHONE_FEATURE` records.

UI/UX Wireframe:
On **Session Prep**, a "Phenotype Snapshot" card might appear, e.g.:

```
+------------------------------------------------+
| PHENOTYPE TRENDS (Past 14 days)                |
| Location:            ███████████████  [18%↓]   |
| Social Calls:        ██████████  [30%↓]         |
| Screen Time:         ███████████  [12%↑]        |
| (Tap for details and timeline)               |
+------------------------------------------------+
```

Clicking "details" opens a modal with graphs (mermaid timeline could show high-level):



This lets the clinician **see trends** and drill into raw data if needed.

AI Components & Prompts: Raw sensor data is pre-processed into features by custom code (e.g. daily average movement, sleep duration). Then an AI task may trigger on new features: e.g. “Detect significant drop in any metric.” We define an LLM-free or LLM-aided pipeline: For example, a prompt like: “Patient has X trend in [steps]. Is this clinically significant?” to generate summary text. More directly, a rule-based check (e.g. 20% drop) can generate “Steady Noticed: Step count down 30%.” If needed, an LLM can help contextualize: *“Decreased movement might indicate increased depression.”* The output is a **Steady Noticed** suggestion (Level 3 inference) – not stored as fact until clinician approves.

Security/Consent/Tenancy: All data collection is opt-in. User consent forms must be updated to include passive data. Data is encrypted in transit and at rest. On the backend, treat phone data as highly sensitive – tag events so they never leave Steady (no external sharing) and enforce strict RLS to ensure only that patient’s clinicians see it. No data is sold or used beyond clinical care. The companion app must respect OS privacy rules (ask for location, motion, etc).

Implementation Phases:

1. **Phase 0 – Research & Prototype:** (1–2 weeks) Build minimal phone data collector. Prototype ingestion of one sensor (e.g. step count from HealthKit). Validate event storage.
2. **Phase 1 – Core Platform:** (3–4 weeks) Finalize event schema, API endpoint, background ingestion worker, and database tables. Implement two key features (e.g. daily steps and location variance). Start populating a **SmartphoneFeature** table. Unit tests: data correctly stored, RLS enforced. **DoD:** CLI tool can ingest sample JSON and produce expected DB records; Session Prep pulls in features.
3. **Phase 2 – Feature Expansion:** (3–4 weeks) Add remaining sensors (screen unlocks, app usage, call count). Create aggregator jobs to compute weekly trends. AI alerts: encode simple rules (20% change). Add UI cards in Session Prep. **DoD:** Simulated multi-week data should produce the trend graphs correctly; “Steady Noticed” alerts appear when thresholds met. Tests include end-to-end API calls and UI snapshot tests.
4. **Phase 3 – Refinement & AI:** (2–3 weeks) Improve AI logic (LLM prompts or ML model for anomaly detection). Build clinician UI to approve/ignore phenotyping alerts (so they don’t become memory facts unless cleared). Add telemetry on user consent and usage. Beta release under feature flag. **DoD:** 90% accuracy of alerts on test scenarios; product telemetry shows correct events.

Estimated effort: **Medium (~8–12 weeks)** (multiple sprints).

2. Wearable Device Integration (Physiological Tracking)

Summary: Extend Steady into the realm of physiological monitoring by syncing with patient's smartwatches or fitness bands. Wearables can continuously measure **heart rate, heart-rate variability (HRV), skin conductance, sleep quality, step/activity count**, etc. These are well-known mental-health biomarkers 【23†L25-L34】. For example, low HRV is linked to chronic stress and depression, poor sleep quality often precedes depressive episodes, and increased post-session heart rate may signal high anxiety. By ingesting wearable data (with consent), Steady gains objective data streams that augment self-reports. These feed into Session Prep ("Average nightly sleep 6h, down from 7h"), Response Fingerprint (e.g. "Grounding today brought HR down 15 bpm"), and Therapeutic Load (e.g. "High post-session HR indicates heavy load").

Clinical Value: Wearables turn therapy into a 24/7 collaboration. Clinicians get **real-time biofeedback**: they can see if relaxation techniques actually reduce stress (via HR drop), or if exposure practice spikes arousal excessively. Studies show wearable data improves engagement (patients feel "seen") and adherence 【23†L37-L40】. Sleep and activity are especially critical: "Sleep disturbances are a hallmark of depression, and wearables can monitor sleep duration and quality"【23†L31-L34】, enabling Steady to detect vicious cycles early. Over time, patterns emerge (e.g. "patient's anxiety peaks correlate with elevated resting heart rate and shorter sleep periods"). This can guide personalized interventions: e.g. add a bedtime relaxation if sleep is low.

Data Sources & Provenance: Patients connect their device via HealthKit (iOS) or Google Fit/Bluetooth on Android. We ingest:

- **Heart rate & HRV:** raw BPM data, or nightly average HRV. (Derived from inter-beat intervals.)
- **Sleep metrics:** total sleep time, sleep efficiency, REM vs deep sleep.
- **Activity:** step count, distance, activity intensity minutes.
- **Stress indicators:** some watches measure skin conductance or "stress score".

Each reading is an event: e.g. `WearableDataEvent {patient_id, timestamp, metric, value}`. We version our schema per device (Fitbit vs. Garmin). We retain raw readings for provenance, then calculate daily summaries.

Architecture & Integration:

- **Ingestion:** When patient authorizes, Steady pulls from HealthKit/Google Fit APIs or listens via companion. A background job bundles data and calls Steady's Wearable API (or writes to database if running server-connected).
- **Event Schema:** `WearableReading(patient_id, timestamp, type, value, device_id)`. Metrics: HR, HRV, sleep duration, steps, etc.
- **Processing:** A nightly job computes `DailyWearableFeature` records (e.g. "avg_hr_last24h", "sleep_lastnight_hours", "steps_today").
- **AI Analysis:** Compare daily values to personal baselines. If a biometric is an outlier (e.g. HRV dropped 30%), raise a *Steady Noticed* insight. The AI task might use rules or a trained anomaly detector.
- **Session Prep:** Incorporate key vitals: e.g. "Last night's sleep was 5h (patient baseline 7h)." or charts of HRV over time.

- **UI:** A new “Vitals” or “Wearables” tab on the patient dashboard shows graphs (could use mermaid line charts) or key stats. The Command Center can also list “High stress this morning (HRV low)” under actionable items.

Database Schema:

```
erDiagram
    PATIENTS ||--o{ WEARABLE_DEVICE : owns
    WEARABLE_DEVICE {
        string id PK
        uuid patient_id FK
        string device_type  // e.g. "AppleWatch", "Fitbit"
        string token       // OAuth or device identifier
    }
    PATIENTS ||--o{ WEARABLE_READ : has
    WEARABLE_READ {
        string id PK
        uuid patient_id FK
        datetime timestamp
        string metric   // e.g. "heart_rate", "hrv", "sleep_hours"
        double value
    }
    DAILY_WEARABLE_METRIC ||--|| PATIENTS : for
    DAILY_WEARABLE_METRIC {
        string id PK
        uuid patient_id FK
        date date
        string metric_type // e.g. "avg_hr", "total_steps", "sleep_hours"
        double value
    }
```

- **WEARABLE_DEVICE:** Patient’s linked devices.
- **WEARABLE_READ:** Raw sensor readings (indexed by patient & timestamp).
- **DAILY_WEARABLE_METRIC:** Aggregated daily summaries (unique on patient_id+date+metric_type). Indexed similarly.

Event/API Contracts:

- *WearableDataEvent:* `{patient_id, timestamp, metric, value, device_type}`. Generated when data is pushed.
- *DailyWearableSummaryEvent:* `{patient_id, date, metric_type, aggregate_value}` (e.g. “sum” or “avg”).

API Examples:

- `POST /api/wearable/data` – body: `{"patient_id":"...","readings":[{"ts":"2026-09-02T22:00Z","metric":"sleep_hours","value":6.5}, ...]}`.

- `GET /api/patient/{id}/wearables?from=2026-09-01` – returns recent daily metrics and trends.

UI/UX:

- **Patient Profile Tab:** Show “Vital Signs”. Example dashboard:

```
+----------------------------------------+
| VITAL SIGNS                            |
| Avg Resting HR (24h): 70 bpm (↑5 from baseline) |
| HRV (24h avg): 25 ms (↓10 from baseline)     |
| Sleep last night: 5.5 hr (norm 7.0 hr)       |
| Steps today: 3000 (goal 5000)                |
| [View full trends charts]                   |
+----------------------------------------+
```

- **Session Prep:** Include wearables summary line: *“Last 3 nights: average 6h sleep/night, down from baseline 7h.”*
- **Command Center Row:** If an event triggers (e.g. “sleep <4h 3 nights”), row reason might be “Poor sleep” with detail from wearable data.

AI & Models: The AI Gateway can run a lightweight ML model to detect physiologic stress patterns. For example, a model that ingests HRV, HR, and sleep history to output a *stress index*. Alternatively, use an LLM prompt: “Patient’s HRV dropped 30% and sleep fell to 5h for 2 nights. What does this indicate?” to generate narrative. But primarily, first implementation is rule-based. Future: train a model on historical data linking wearable metrics to symptom scores. The AI outputs are always labeled “Model-based inference” and must be reviewed by clinician (unless self-evident like HRV).

Security/Consent: Patients must explicitly link a device. OAuth or code linking ensures consent. Only clinicians of that patient (and tenant admins) see the data. Wearable streams are stored in HIPAA-compliant encrypted storage. Data is only used for care – no analytics or third-party sharing. If a patient unlinks a device, past data is retained but no new data ingested.

Implementation Plan:

- **Phase 0 – Prototype Sync (2w):** Integrate with one platform (e.g. HealthKit). Set up data flow, basic DB, and test ingestion of HR and sleep.
- **Phase 1 – Core Schema & Events (4w):** Finalize tables/events above. Build full ingestion pipeline, error handling (e.g. token refresh). Add UI placeholders and test with mock data. **DoD:** Wearable metrics correctly stored for real/pseudo user; UI shows example data; events logged.
- **Phase 2 – Feature Expansion (4w):** Add additional metrics (steps, HRV), build daily summary job, connect to existing sessions. Implement “last night’s sleep” in Session Prep. Unit tests for data quality and dashboard.
- **Phase 3 – Analytics & AI (3w):** Implement simple alerts (e.g. “HRV < threshold → flag”). Build a basic linear model or use rules. Add clinician toggle to acknowledge or override. Develop tests using synthetic anomalies.

- **Phase 4 – Polishing (2w):** Support multiple wearable types; add device linking UI/UX; finalize documentation.

Estimated effort: **Large (~12–15 weeks)** due to device integration complexity.

3. Medication Adherence Tracking

Summary: Many mental health conditions require daily medications, and **non-adherence is a major cause of relapse**. Steady can help by letting clinicians enter a patient's medication regimen (names, dosages, schedule) and then having the patient confirm intake via the Companion app or daily check-ins. The system logs "medication taken" events. It can send reminders (push notification or SMS) at scheduled times. The Command Center flags patients who miss doses (e.g. 2 days in a row), and Session Prep can display adherence rate. This turns Steady into a *medication support tool*.

Clinical Value: Providers often cannot distinguish whether treatment failure is due to the medication itself or the patient not taking it【31†L150-L159】. With real adherence data, clinicians make better decisions (adjust doses vs. adjust treatment approach). Early prompts on missed meds allow immediate intervention: e.g. nurse or care manager can outreach. Studies show reminder systems significantly improve adherence across conditions【30†L1-L4】. In depression care, correlating mood changes with med-taking can reveal if side effects are causing dropout. As HealthArc notes, "RPMs make it possible to track symptoms and correlate them with medication use"【31†L156-L164】.

Data Sources & Provenance: Data is patient-reported (or confirmed by app). Two data streams:

- **Scheduled regimen** (Level 2 memory): events like `MedicationScheduleAdded {patient, med_name, dose, times}`. Clinician or patient can enter this.
- **Intake logs** (Level 1 or 2): Each time the patient takes a dose, an event `MedicationTaken(patient, med, timestamp)` is recorded via companion app or web portal. If skipped, patient can optionally log a reason (side effects, forgot, etc). Unconfirmed intervals can be inferred "missed dose" events by logic.

Provenance: mark these as patient self-report. They become part of the patient's memory after clinician review.

Architecture & Integration:

- **Database:** New tables `MedicationSchedules` and `MedicationEvents`.
- **Command Center:** New sub-queue "Medication Alerts". Patients with two or more missed doses trigger "Needs Attention: Medication Adherence".
- **Session Prep:** Show adherence percentage (e.g. "8/10 doses taken since last visit") and any patterns ("weekends are worse"). Also list new meds started.
- **AI:** Correlate adherence with outcome: e.g. if PHQ9 worsened after missed doses, Steady can note "Symptoms rose when meds were missed." This is a *Steady noticed* hypothesis, not auto-applied.

Data Model:

```
erDiagram
    PATIENTS ||--o{ MEDICATION_SCHEDULE : has
    MEDICATION_SCHEDULE {
        string id PK
        uuid patient_id FK
        string medication_name
        string dosage
        string frequency  // e.g. "2x/day", "every morning"
        bool active
    }
    PATIENTS ||--o{ MEDICATION_EVENT : has
    MEDICATION_EVENT {
        string id PK
        uuid patient_id FK
        string medication_id FK  // references MEDICATION_SCHEDULE.id
        timestamp taken_time
        string status   // "taken", "skipped"
        string note     // e.g. "felt nauseous"
    }
```

- **MEDICATION_SCHEDULE:** One per med per patient.
- **MEDICATION_EVENT:** Each dose taken or explicitly skipped (status). Index on patient and medication_id, and time.

Event/API Contracts:

- *MedicationScheduleEvent:* `{patient_id, medication_id, name, dosage, frequency}` when schedule created/edited.
- *MedicationTakenEvent:* `{patient_id, medication_id, timestamp, status}`.

API:

- `POST /api/patient/{id}/medication/schedule` – adds or updates schedule.
- `POST /api/patient/{id}/medication/log` – e.g. `{"medication_id":"123","time":"2026-09-03T08:00Z","status":"taken"}`.
- `GET /api/patient/{id}/medication/adherence?since=2026-08-01` – returns summary adherence (e.g. “12/14 doses taken, 85%”).

UI/UX:

- **Medication Dashboard:** On patient profile, a “Medications” section:

```
Medication        Dosage     Schedule        Adherence
Lexapro           10mg       Every AM        14/14 (100%)
Wellbutrin XR     150mg      Every PM        12/14 (85%) [Needs attention]
```

Clicking a row shows details and a calendar of taken vs. missed.

- **Patient App:** Notifications at scheduled times; simple “Did you take Lexapro 10mg?” with “Yes/No” buttons. Misses can be logged.
- **Command Center:** A row “Jane D – Medication Alert” with reason “Missed 3 doses of Lexapro in last week” and quick actions (“Send reminder”, “Schedule call”).

AI & Prompts: An AI task can examine `MedicationEvent` and symptom scores. Example prompt: *“Patient reported higher anxiety after missing doses of Medication X. Does this suggest anything?”* The answer (or rule) would be: *“Symptom spike coincides with missed medication, consider side-effect vs non-adherence explanation.”* Again, any suggestion is “Possible inference” for clinician to evaluate.

Security/Consent: Patients must opt into medication tracking. Data is optional and self-reported, so less sensitive than vitals, but still privacy-protected. Clearly state that logs are seen by clinician. Automated reminders are only sent if patient opts in; can turn off at any time.

Implementation Plan:

- **Phase 1 – Data Model (2w):** Create tables and API. Allow clinicians to add medication schedules and patients to log intake. Build basic UI. **DoD:** Can add/edit schedules; events stored properly; adherence query returns correct percent.
- **Phase 2 – Reminders & Alerts (3w):** Implement reminder jobs (push notifications). Define alert rules (e.g. missed 2+ consecutive). Integrate with Command Center. **DoD:** Missed-event logic triggers alerts in test scenarios. UI shows appropriate rows.
- **Phase 3 – Analytics (2w):** Add correlation logic (e.g. compare PHQ vs adherence). UI in Session Prep shows chart of symptoms vs med-taking. Add tests.
- **Phase 4 – Edge Cases (1w):** Handle PRN medications, PRN scheduled differently. Add tests for timezone.

Estimated effort: **Medium (6–8 weeks)**.

4. Digital Therapeutic Modules (Guided Interventions)

Summary: Provide patients with a **menu of evidence-based self-care exercises**—essentially Steady’s own library of digital therapeutics. This could include CBT worksheets (e.g. identifying cognitive distortions), guided breathing/meditation exercises, journaling prompts, or exposure therapy videos. Clinicians can assign modules, and Steady tracks completion and patient responses. These structured tasks complement therapy: for example, a patient with social anxiety might be assigned an “Exposure: Initiate small talk” module with steps. Completing these modules generates new data (e.g. self-reported outcome), which feeds back into Steady’s memory and the Response Fingerprint.

Clinical Value: Clinicians often give “homework” but it’s hard to track. Embedding it in the platform closes the loop. Digital interventions are proven: they increase engagement and outcomes, especially when tailored【41†L1-L4】. Steady’s modules can adapt over time (based on the Response Fingerprint, we “suggest next exercise”). Patients get **real-time feedback** (“Nicely done!”) and see progress bars, which improve motivation【41†L1-L4】【41†L7-L10】. For example, after assigning a mindfulness module, Steady can confirm if the patient’s stress rating dropped. This makes therapy more continuous and data-driven.

Data & Provenance: - **Module Catalog (static content):** Stored on server (title, description, expected outcome). Not patient-specific, but versioned.

- **Assignments (Level 2 memory):** `ModuleAssignment {patient, module_id, assigned_by, due_date, status}` events.
- **Completions (Level 1):** `ModuleCompletion {patient, module_id, timestamp, input_data}`. Input may include the patient's input (e.g. answers to CBT questions, or a journal entry).
- **Outcomes (Level 3):** From completion data, we can derive metrics (e.g. confidence level). These can inform AI insight (e.g. "Patient reports 7/10 anxiety during exposure").

Architecture & Integration:

- **UI:** A "Modules" section in patient view. Clinicians select modules by drag-drop (or search). Assigned modules appear in patient's app as tasks with due dates.
- **Events:** When a module is assigned, fire `ModuleAssignedEvent`. When done, fire `ModuleCompletedEvent`.
- **Session Prep:** List "Last module completed: X" and outcome (e.g. "Felt 8/10 anxious"). In dashboard, show completion rate.
- **AI Gateway:** Use an LLM to triage module difficulty or recommend next steps. For example, prompt: *"Patient reports no improvement after module Y. Recommend alternate approach."* Could also use sentiment analysis on journaling inputs (ties to Voice/Text feature).
- **Memory:** Clinician approves what counts as success. E.g. "Patient's fear ranking decreased from 9 to 5 after module" can become a memory **Intervention Response**.

Database Schema:

```
erDiagram
    PATIENTS ||--o{ MODULE_ASSIGNMENT : has
    MODULE_ASSIGNMENT {
        string id PK
        uuid patient_id FK
        string module_id FK
        timestamp assigned_on
        date due_date
        enum status   // "pending", "completed", "overdue"
    }
    PATIENTS ||--o{ MODULE_COMPLETION : has
    MODULE_COMPLETION {
        string id PK
        uuid patient_id FK
        string module_id FK
        timestamp completed_on
        string input_data  // JSON or text of patient responses
```

```
        double outcome_score // e.g. self-rated symptom change
    }
```

- **MODULE_ASSIGNMENT:** Who was assigned what, with status.
- **MODULE_COMPLETION:** Details of completion. Index on patient and module_id.

Event/API Contracts:

- *ModuleAssignedEvent:* `{patient_id, module_id, clinician_id, due_date}`.
- *ModuleCompletedEvent:* `{patient_id, module_id, timestamp, responses: {...}}`.

API:

- `POST /api/modules/assign` – assign a module to a patient.
- `GET /api/patient/{id}/modules` – list assignments with status.
- `POST /api/modules/complete` – patient submits module (with answers).

UI/UX:

- **Clinician View:** “Add Module” button on patient record. A searchable list of module tiles (with descriptions and expected duration). Once assigned, shows in a “Pending Modules” list with due dates.
- **Patient App:** A “My Tasks” list:

```
[ ] Journal: Gratitude (due Sep 4)   [?]   [Complete]
[ ] Breathing Exercise (due Sep 5)   [?]   [Complete]
```

Clicking “Complete” opens the module: e.g. a multi-step journaling interface with text fields. On finish, prompt “Rate your current anxiety (1-10)”. Then the app logs the completion.

- **Session Prep:** Note “Patient completed Gratitude Journal on Sep 1: reported gratitude feelings 8/10 (from 3/10 before).”

AI & Prompt Contracts: Modules themselves might be static content, but the follow-up can involve AI. For example, after completion: *“Based on the patient’s journal responses, suggest a follow-up comment.”* Or *“Patient still rates anxiety 8/10 after module. Recommend next module.”* These are optional and clinician-facing. Steady should label these “AI Suggestion: [text]” for clinician editing.

Security/Privacy: Module content is by design not highly sensitive (like worksheets). But patient responses (especially journaling) can be private. They should only be visible to that patient and their care team. Data should be encrypted; patients must consent to the therapist reviewing their submitted content.

Implementation Plan:

- **Phase 1 – Module Framework (3w):** Define modules data (hard-code 3–5 initial modules). Create assignment/completion tables and API. Basic clinician UI to assign and patient UI to list tasks. **DoD:** Assignments save correctly; patient can submit one; status updates to “completed”.

- **Phase 2 – Tracking & Session Prep (3w):** Display completion status in Command Center (e.g. "2 tasks pending"). Include module summaries in Session Prep. Tests to simulate user completing or missing.
- **Phase 3 – AI Insights (2w):** Create LLM prompts for recommendation after completion (optional). Add "Steady Noticed" flags if a module's goal not met (e.g. "No drop in anxiety after exposure module").
- **Phase 4 – Content Expansion (2w):** Add more modules, allow video or interactive content. Ensure multi-language support if needed.

Estimated effort: **Large (8–10 weeks)** (due to UI and content requirements).

5. Voice and Sentiment Analysis (Vocal Biomarkers)

Summary: Use AI to analyze patients' speech and text for affective signals. Advances in vocal biomarker research show that **short voice samples can reveal depression, anxiety, and cognitive state**【39†L425-L434】. For example, researchers correctly identified moderate-to-severe depression from **25-second smartphone speech samples with ~70% accuracy**【39†L432-L437】. Similarly, changes in word usage (sentiment, lexical complexity) correlate with mood. Steady can allow optional voice journaling (patient speaks into the app) or analyze speech during video calls (if recorded/allowed). Each audio recording is sent to an AI model (through the AI Gateway) that outputs a **Depression/Anxiety Score** and sentiment rating. For text (e.g. chat with Companion or journal entries), NLP models output emotion metrics ("sadness", "negativity"). These become part of the patient's memory and prompt alerts.

Clinical Value: This adds an *objective lens* to what patients say. For example, a patient might report "fine," but their monotone voice and slow speech (reduced energy) could indicate low mood. Automated voice analysis can pick up subtle cues therapists might miss【39†L425-L434】. Tracking sentiment over time yields a quantitative trend. Clinicians get a "Sentiment Meter" alongside session notes, and Steady can preemptively flag warning phrases ("I want to give up") or vocal stress (via LLM classification) as part of safety monitoring. This enhances traditional check-ins with data-driven mental state indicators.

Data & Provenance: Data comes from two possible sources:
1. **Recorded Sessions or Voice Journals:** Patient-recorded audio snippets (with timestamp). Mark as highly sensitive.
2. **Text Logs:** Transcripts from AI Companion conversations or any in-app writing (journals, messages).

Provenance: These are **Level 1: raw patient data**. We then apply AI (LLM or ML) to extract emotional features (Level 2 memory) or risk scores (Level 3 inferences). For example, after transcription: `VoiceAnalysisEvent {depression_score:0.8, anxiety_score:0.3}`.

Architecture & Integration:

- **Ingestion:** Patients can optionally record a 1–2 minute voice memo in the app. On submission, Steady sends the audio to the AI Gateway. Alternatively, companion chatbot voice notes can auto-process.
- **AI Processing:** Use specialized models (e.g. open-source speech emotion recognizer) or LLMs with prompts. For instance, the AI Gateway receives: *"<30s audio> user is a 30y female complaining of...;*

Analyze for depression/anxiety indicators." The prompt could instruct the model to output a JSON with scores.

- **Results:** The output `VoiceSentimentResult` is stored (with links to source). The system compares to prior results to find trends.
- **UI:** Show a small "Mood Trend" widget on Session Prep (e.g. a sparkline of sentiment over time) or an icon (😊😐😟) in the patient's row. The clinician can click to see underlying analysis (e.g. "Spoke softly, little variation in pitch – scored high depression features").

Data Model:

```
erDiagram
    PATIENTS ||--o{ VOICE_RECORDING : has
    VOICE_RECORDING {
        string id PK
        uuid patient_id FK
        datetime timestamp
        string transcript  // if stored
    }
    VOICE_ANALYSIS ||--|| VOICE_RECORDING : analyzes
    VOICE_ANALYSIS {
        string id PK
        string recording_id FK
        timestamp analyzed_on
        double depression_score   // 0.0–1.0
        double anxiety_score
        double valence_score      // general positivity/negativity
        string summary           // optional text snippet
    }
```

- **VOICE_RECORDING:** Raw audio reference (actual files stored securely, not in DB). Can include transcript if doing speech-to-text.
- **VOICE_ANALYSIS:** Results from ML/AI. Indexed by recording.

Event/API:

- *VoiceRecordingEvent:* `{patient_id, recording_id, timestamp, transcript}`.
- *VoiceAnalysisEvent:* `{recording_id, depression: #, anxiety: #, summary: "..."}`.

API:

- `POST /api/voice/record` – patient uploads audio.
- `GET /api/patient/{id}/sentiment-trend` – returns list of analysis scores over time.

UI/UX:

- **Session Prep:** A "Vocal Mood" card with small emojis or gauge: e.g. "Mood: 😟 (-0.6)" (where -1 to +1 sentiment). Or a timeline sparkline of average sentiment.

- **Patient History View:** List of voice entries with analysis: “Sep 1: score 0.3 (some depression features), ‘I feel quite tired’”.
- **Command Center:** If voice analysis crosses risk threshold (e.g. score >0.7 depression), row marked “Needs Attention: High vocal distress.”

AI Components: Use a combo of ML models. For voice: libraries like [pyAudioAnalysis]() or [affect detection models]() to extract acoustic features, then feed to a classifier. For simplicity, start with an open-source API or LLM prompt (e.g. OpenAI with a carefully crafted prompt: *“Analyze this speech excerpt and return scores.”*). For text: use a sentiment analyzer (or LLM with `classify sentiment` prompt). Define strict prompt/response schema (JSON). These outputs are **Level 3 inferences** (AI), clearly labeled. Clinicians see them as “AI suggestion: patient sentiment appears strongly negative.”

Security/Consent: Very sensitive! Recording sessions or journals requires explicit consent. All processing is HIPAA-compliant; transcripts and audio are encrypted and deleted on request. Only care team sees analysis. No audio is stored longer than necessary (prefer process-and-delete, or store only after anonymization).

Implementation Plan:

- **Phase 1 – Text Sentiment (2w):** Before voice, implement sentiment analysis on patient text (cheaper). Integrate an open-source sentiment API or use LLM. Show sentiment score for check-in texts. **DoD:** Known test inputs yield correct polarity.
- **Phase 2 – Voice Recording Setup (3w):** Add UI to record/submit audio. Implement backend storage. Use speech-to-text to get transcript. Call AI to get analysis. **DoD:** Upload a sample audio, receive JSON with plausible scores (mock model).
- **Phase 3 – Refinement (3w):** Tune the AI model with fine-tuning or thresholds. Build UI cards. Add tests with known audio samples (public datasets). Check RLS.
- **Phase 4 – Integration (2w):** Connect analysis to Command Center and Session Prep. Allow clinicians to disable if needed.

Estimated effort: **Medium (6–8 weeks)**.

Feature Comparison & Roadmap

The above features build on each other. Figure 1 outlines their phased implementation order (Phase 0 is our existing notes system):

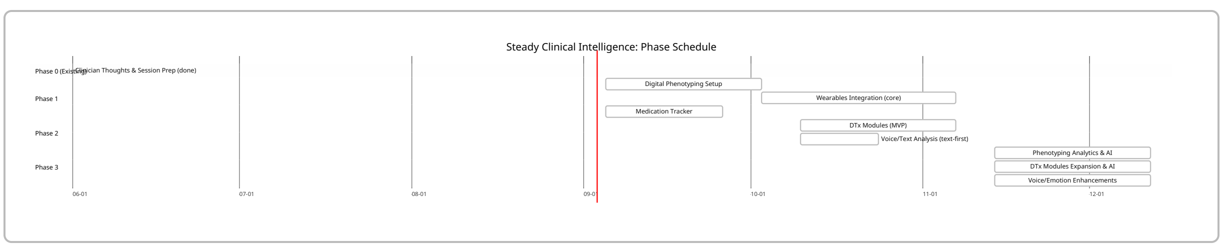


Table 2. Summary of Features. Each feature tightly reuses Steady’s multi-layer memory model, AI Gateway, TenantContext and security framework:

Feature	Clinical Impact	Key Dependencies	Est. Effort (wk)	Priority
Passive Phenotyping	Early detection via behavior patterns【17†L112-L121】	Smartphone SDK, Companion App, AI models	8–12	High
Wearable Integration	Continuous vitals/sleep tracking【23†L25-L33】	Device APIs, new event schemas	12–15	High
Medication Tracking	Improve adherence, interpret treatment【31†L150-L159】	Scheduling model, notifications	6–8	Medium
Therapeutic Modules	Extend therapy between sessions (CBT, etc.)【41†L1-L4】	Content library, UI, AI recommendations	8–10	Medium
Voice/ Sentiment AI	Objective mood markers from voice/text【39†L425-L434】	Audio/text ingestion, AI models	6–8	High

Each feature enriches Steady's longitudinal data. By implementing them in the above order, engineering can reuse platform primitives (the event spine, session prep, command center, AI Gateway) and ensure smooth integration. The result will be a fully preventive, data-driven behavioral health platform in line with cutting-edge practice【14†L175-L182】【31†L230-L238】.
